

Transfer Manifest - Technology #07

Self-Powered Sensing via Piezoelectric Energy Harvesting

YOU RECEIVE:

- + Self-powered sensing concept + R1 correction (duty-cycled architecture)
- + Piezo harvesting governing equations + power budget model
- + Bench test protocols (power, fatigue, biocompatibility)
- + External precedent catalog (PVDF, PZT, lead-free piezo, harvesting ICs)
- + Critical UNKNOWN disclosure (strain amplitude, chronic depoling, biocompatibility of piezo materials)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production piezo transducer design (material, geometry, integration with catheter)
- > Power conditioning circuit (production PCB, miniaturized)
- > Sensor selection + integration + self-referencing architecture
- > Manufacturing process for catheter-integrated piezo + circuit + sensor
- > Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2)
- > Clinical validation (accuracy + longevity)
- > Long-term reliability testing (fatigue, depoling, biocompatibility)

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

NOT AVAILABLE:

- Physical prototype
- Clinical data
- Qualified supplier
- Granted IP
- Independent engineer evaluation

This manifest is honest. Items marked 'not available' do not exist.